Eugen Miknevic

AI Engineer for Casting & Manufacturing Simulation

Physics-ML • GNN Surrogates • GPU Solvers • 20 Years Foundry Engineering

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PROFESSIONAL SUMMARY

AI engineer with 20 years of foundry and casting engineering behind the models. I build physics-based machine learning and simulation software for manufacturing end to end: a GPU solidification solver validated against commercial simulation, an open-weights GNN surrogate trained on 14,937 FEM simulations, and human-in-the-loop LLM systems piloted on the shop floor. Former Head of Design & Metallurgy (400+ casting technology packages) and lead of greenfield foundry projects; expert user of MAGMASOFT, ProCAST and WinCast.

TECHNICAL SKILLS

Physics-ML & ML: Graph neural networks (MeshGraphNets), PyTorch, PyTorch Geometric, surrogate modelling, gradient boosting, scikit-learn, TensorFlow, CNNs

Simulation & HPC: FEM / CVFEM, enthalpy solidification, transient heat transfer, implicit solvers (PCG), CUDA.jl, FEniCSx, Gmsh, V&V against commercial codes

LLM & agents: RAG (pgvector, ChromaDB), structured output, AI agents and tool use, human-in-the-loop design, on-prem LLM / STT (Ollama, Qwen, Whisper), OpenAI / Claude / DeepSeek APIs, LangChain

Software: Python, Julia, TypeScript, SQL; FastAPI, React, SvelteKit, three.js; PostgreSQL, Redis; Docker, GitHub Actions, Linux; pytest, Playwright; TDD, ADRs, agentic coding (Claude Code, Codex)

CAE & casting: MAGMASOFT, ProCAST, WinCast, SolidWorks, CadQuery; gating and feeding design, defect analysis, investment, sand, no-bake, lost foam, die casting, V-process, 3D-printed moulds

Languages: German (C1), English (B2, professional working), Ukrainian (native), Russian (fluent), Polish (B2)

KEY AI & SIMULATION PROJECTS

FoundryFlash: GPU Casting Simulator | *Julia, CUDA.jl, Gmsh, FastAPI, React, three.js* **2026**

- Wrote a 3D CVFEM enthalpy solidification solver with CPU and CUDA backends; solves a 1.6M-element case in 9.9 s on GPU vs 44.9 s on 8 CPU threads. Cut meshing 14× (57 s to 4 s) and job start-up from 27 s to under 0.1 s with a persistent solver daemon.
- Validated against an independent FEniCSx reference (solid-fraction MAE 0.0086) and against commercial simulation on production parts; a sensitivity study traced the gap to mould thermal properties and closed it from +51% to within 3.7%.
- Browser workflow from STEP upload to meshing, material setup, 3D field viewer, feeder sizing and PDF report; about 7,500 automated tests across Julia, Python and TypeScript.

castsolid-gnn: Solidification Surrogate | *PyTorch Geometric, FEniCSx, Gmsh* **Hugging Face | GitHub | 2026**

- Trained a hybrid GNN + gradient-boosting surrogate (12 message-passing blocks, FiLM conditioning, 1.85 M parameters) on 14,937 FEniCSx simulations; predicts solidification fields and hot spots from a STEP file in seconds, with hot-spot overlap 0.917 and nodal MAE 4.26 s on unseen geometry families.
- Published weights and inference code under Apache-2.0, verified tensor-by-tensor and on 10,002 probe rows against the internal checkpoints; shipped CAD healing, meshing, out-of-distribution checks, shrinkage screening (Niyama proxy, feeder modulus) and a browser viewer around the model.

GussCore: AI-Assisted Foundry ERP/MES | *FastAPI, SvelteKit, PostgreSQL, Ollama* **GitHub | 2026**

- Workers report by voice or photo through on-prem Whisper and Qwen LLM/VLM; every AI proposal is confirmed by a person before it is saved. Hexagonal architecture, event-sourced audit, 300+ tests, 15 ADRs.

AlloyForge: Superalloy Design Assistant | *FastAPI, pgvector, bge-m3, pycalphad* **GitHub | 2026**

- PHACOMP / CALPHAD screening of heat-resistant superalloys with RAG over metallurgy literature.

PROFESSIONAL EXPERIENCE

Founder & AI Engineer, FoundryFlash **May 2026 – Present**
Independent R&D *Senftenberg, Germany*

- Built FoundryFlash single-handed, from numerical core to web product; about 1,500 commits in 12 months.
- Designed and built GussCore for a two-plant foundry group; piloted at both plants.
- Built a B2B lead-generation agent (EU TED tender monitoring, lead qualification, outreach drafts) whose GDPR and UWG rules are enforced in deterministic, fail-closed code rather than by the model; 275 tests.

Foundry Engineer, Project Management **Sep 2022 – Apr 2026**
SLR-Elsterheide GmbH *Elsterheide, Germany*

- Built a RAG knowledge system over customer standards documentation, in daily use by the engineering team; wrote Python tools that automated product-image processing and drawing-revision comparison.
- Designed gating and feeding in MAGMASOFT; introduced 3D printing in the pattern shop, cutting tooling lead times several-fold.
- Led technical talks with international customers, turning requirements into simulation-ready specs.

Foundry Engineer, Project Management **Aug 2020 – Aug 2022**
Kunstgiesserei Lauchhammer KG *Lauchhammer, Germany*

- Led commissioning of a modern induction melting complex and investment casting line, concept to production.
- Designed equipment that halved cycle time and cut energy use by 30%; built a large-format wax 3D printer.

Head of Design & Metallurgy **Mar 2015 – Jul 2020**
Giesserei Elsterberg GmbH *Elsterberg, Germany*

- Delivered 400+ complete casting technology packages (CAD, gating and feeding, WinCast simulation, process parameters) for complex industrial components; led design, metallurgy and production as one team.

Foundry Projects (alongside employment) **2016; 2020**
Karlskrona LC AB; Hidrolats SIA *Kazakhstan; Latvia*

- Commissioned a greenfield foundry for uranium-mining pump systems (Kazakhstan); developed FEM-supported casting technology for diesel-engine castings for railway and defence (Latvia).

Foundry Technology Engineer **Nov 2010 – Aug 2014**
Physico-Technological Institute of Metals and Alloys; Grand Steel Foundry *Kyiv / Malyn, Ukraine*

- Developed lost-foam technology for metro tunnel-lining segments, about 20% cheaper than no-bake moulding; commissioned a V-process moulding line for railway castings through to stable serial production.

PUBLICATIONS

- Miknevič, E. "A Faster First Look at Casting Solidification." *Foundry Management & Technology*, Sep 2026.
- Miknevič, E. "Certainty in the Early Design Stages of Foundry Engineering." *Foundry-Planet*, Sep 2026.

EDUCATION & CERTIFICATIONS

Master's Degree, Foundry Engineering **2003 – 2009**
Igor Sikorsky Kyiv Polytechnic Institute; ZAB: equivalent to a German engineering master's (2020) *Kyiv, Ukraine*

- **Professional title "Ingenieur"**, granted by Ingenieurkammer Sachsen, Sep 2020; entered in the register under § 34 SächsIngG.
- **AI Engineering**, Turing College, Sep 2026: 520 hours; LLM applications, LangChain, RAG, AI agents, capstone.
- **A Deep Dive into Claude Code**, Profile Virtual School, Jun 2026; **WandelBAR**, FSR@BB, Mar 2026.